Chapter 5 exercises. Functions, counting, and cost proofs

1 • One formula with different types

For $f : \mathbb{Z} \to \mathbb{Z}$, $f(n) = 3n - 7$, determine injectivity, surjectivity, and invertibility, with explicit quantified witnesses. Change only the codomain to make a bijection and give its inverse. Then change both domain and codomain to rationals and decide again.

2 • Nonempty fibers and named servers

Send five labeled nodes to three named servers, with every server receiving a node. Use the Stirling recurrence to derive needed values and count assignments. Check using alternating weights for missed servers. Why does the answer differ with unnamed servers?

3 • A finite-set equivalence

Prove a self-map of a finite set is injective iff onto, including the empty set. Give natural-number self-map counterexamples to each direction for infinite sets.

4 • Can composition hide collisions?

Prove $g \circ f$ injective iff f and $g|_{f(A)}$ are injective. Give explicit sets with injective composition but noninjective g. Give onto composition with f not onto, and identify an additional condition restoring that inference.

5 • What pigeonholes guarantee

For 23 integers modulo 22, how many congruent integers are guaranteed? Show three cannot be guaranteed. For $k \geq 1$ boxes, find the least total forcing $t \geq 1$ objects in some box, proving sufficiency and giving a counterexample with one fewer object.

6 • Numeric value and bit length

For integer $N \geq 1$, repeatedly replace it by $\lfloor N/2 \rfloor$ until zero. In terms of initial bit length b, find exact iterations. Analyze fixed-word representation, copying remaining bits at every step, and adjusting a read boundary. State costs excluded from each model.

Chapter 5 exercises. Functions, counting, and cost proofs (continued)

7 • The full criterion for a quotient formula

For positive m, n, study $h : \mathbb{Z}/m\mathbb{Z} \to \mathbb{Z}/n\mathbb{Z}$, $h([a]_m) = [a]_n$. Prove well-definedness iff $n \mid m$. In the valid cases give surjectivity and injectivity criteria, and list fibers for $m = 6, n = 3$.

8 • Structural counting with partitions

Prove every partition can be the nonempty fibers of a function, with equal-value relation exactly the partition's equivalence relation. For domain and codomain sizes m, n, derive the count with image size k, and the identity summing over all k. Check $m = 4, n = 3$.

9 • Identities and operation tables

For a set of size $n \geq 1$ and specified e, count closed binary operations with two-sided identity e, then those also commutative. Why can requiring some identity be counted by multiplying by n? Evaluate at three. Construct an operation where every element is a left identity but no two-sided identity exists.

10 • Keep the correct objects in existence proofs

Prove any seven integers have a nonempty contiguous subsequence whose sum is divisible by seven. Why are residues of only the seven original values insufficient? Prove 17 distinct reals contain a strictly increasing or decreasing subsequence of length five, and give a sixteen-term counterexample to that guarantee.

11 • Proving binary exponentiation

Prove the lecture's program initialized by $r = 1, b = a, e = N$ correct with associative multiplication and an identity, including $N = 0$. Trace 2^{11}, recording e, b, r and updated r. Give the general multiplication count and explain why it does not prove logarithmic bit time for arbitrary integers.

Chapter 5 exercises. Functions, counting, and cost proofs (continued)

12 • Witnesses for growth and averages

Prove $7n^2 + 2n + 1 \in \Theta(n^2)$ with constants, and negate the quantifiers to prove it is not $O(n)$. A length-five linear search has absence probability $1/4$ and conditionally uniform successful positions. Find expected comparisons. If all successes are instead at the first cell, still with success probability $3/4$, is the expectation unchanged?

References and attribution

Ralph P. Grimaldi. Discrete and Combinatorial Mathematics: An Applied Introduction, 5th Edition, 2004.

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https://github.com/xinhi6716/computer-science-self-study

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